14/5/2025(E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025 Maximum Marks: 100 Examination: ESE Examination Duration:3 Hrs.		
Programme code: 05 Programme: B.Tech Mechanical Engineering	Class: ~~FY/SY~~/TY/~~LY/MTECH~~	Semester:_VI_(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: ~~COMP/ETRX/EXTC/IT~~/MECH/~~RAI/AIDS/CCE/EXCP~~	
Course Code: 116U05E616	Name of the Course: Operation Research	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Two**	**20**
i)	Consider a 3 machine and 5 job flow shop problem as shown in Table given below	10

Job	Processing time		
	Machine 1	Machine 2	Machine 3
1	10	10	12
2	12	8	20
3	16	6	14
4	12	4	20
5	20	8	8

a) Check whether the Johnson's algorithm can be extended to this problem.
b) If yes, find the optimal sequence and the corresponding makespan

ii)	Consider a two 'jobs and m machine job shop scheduling' problem which is presented in Table below. Use graphical method to minimize the time needed to process the following job on the machines shown (i.e. for each machine find the job which should be scheduled first). Also, calculate the total elapsed to complete both jobs.	10

Job 1		Job 2	
Sequence	Time (hrs)	Sequence	Time (hrs)
A	5	B	7
B	6	C	6
C	4	A	5
D	8	D	4
E	4	E	8

Que. No.	Question	Max. Marks
iii)	Consider the following problem in single machine scheduling with independent jobs.	10

Obtain the optimal schedule for each of the following performance measures:

a) To minimize mean flow time
b) To minimize the maximum lateness
c) To minimize weighted mean flow time

Job, j	t_j	d_j	w_j
1	5	10	2
2	12	16	1
3	8	11	1
4	10	16	2
5	3	6	3
6	15	25	4
8	8	12	2
9	6	14	3

Que. No.	Question	Max. Marks
Q2	Solve the following	20

Consider the transportation problem shown below

		Market 1	2	3	4	5	Supply
Plant	1	10	2	16	14	10	300
	2	6	18	12	13	16	500
	3	8	4	14	12	10	825
	4	14	22	20	8	18	375
Demand		350	400	250	150	400	

Find the initial basic feasible solution using each of the following methods and compare their total costs.

a) Northwest corner method
b) Least cost cell method
c) Vogel's approximation method

OR

a) Solve the following assignment problem and obtain the minimum cost at wich all jobs can be performed **10**

Worker	Job (Cost in 00Rs) 1	2	3	4	5
A	25	18	32	20	21
B	34	25	21	12	17
C	20	17	20	32	16
D	20	28	20	16	27

b) A company plans to assign 5 salesmen to 5 districts in which it operates. Estimates of sales revenue in thousands of rupees for each salesman in different districts are given below in following table. In your opinion what should be th placement of the sales men if the objective is to maximise the expected sales revenue. **10**

Sales man	District D_1	D_2	D_3	D_4	D_5
S_1	40	46	48	36	48
S_2	48	32	36	29	44
S_3	49	35	41	38	45

Que. No.	Question	Max. Marks

S_4	30	46	49	44	44
S_5	37	41	48	43	47

Q3 Solve any **Two** — **20**

i. Players A and B play a game in which each player has three coins (20p, 25p, and 50p). Each of them selects a coin without the knowledge of the other person. If the sum of the values of the coins is an even number, A wins B's coin. If that sum is an odd number, B wins A's coin. — 10

a) Develop a payoff matrix with respect to player A.

b) Find the optimal strategies for the players.

ii Consider the 4X4 game played by players A and B shown in table below and solve it optimally. — 10

		Player B			
		1	2	3	4
Player A	1	6	2	4	8
	2	2	-1	1	12
	3	2	3	3	9
	4	5	2	6	10

iii Solve the following 3X5 game using dominance property shown below — 10

		Player B				
		1	2	3	4	5
Player A	1	2	5	10	7	2
	2	3	3	6	6	4
	3	4	4	8	12	1

Q4 Solve any **Two** — **20**

i. A 4-ton vessel can be loaded with one or more of three items. The following table gives the unit weight, w_i, in tons and the unit revenue in thousands of dollars, r_i, for item i. How should the vessel be loaded to maximize the total return? — 10

Item i	w_i	r_i
1	2	31
2	3	47
3	1	14

Because the unit weights w_i and the maximum weight W are integer, the state x_i assumes integer values only.

ii. Four types of items are under consideration for loading on a plane, with an unlimited supply of each type. The weights and values of various types of items are given below. — 10

Item	Weight (tons)	value
A	3	22
B	6	44
C	5	35
D	4	24

Determine as to which items should be loaded on the plane, and what quantities, if the maximum capacity of the plane is 13 tons and the objective is to maximize the value of the shipment. Give a mathematical representation of the problem and solve it by the dynamic programming.

iii. A company has trained five salespersons in the past three months, whom it — 10

Que. No.	Question	Max. Marks
	wants to assign to three sales regions: north, West, and South. It has been estimated that increases in sales (in lakhs of rupees) over next year in each region, if each were allocated different number of salesmen are as indicated below.	

Salespersons	Region		
	North	West	South
0	0	0	0
1	61	36	66
2	86	61	86
3	106	91	96
4	121	126	116
5	136	166	146

Q5	Solve any **four**	**20**
	i) A firm is considering replacement of an equipment whose first cost is Rs. 4000/- and the scrap value is negligible at the end of any year. Based on experience, it has been found that the maintenance cost is zero during the first year and it is Rs. 1000/- for the second year. It increases by Rs. 300/- every year thereafter. a) When should the equipment be replaced if $i = 0\%$ b) When should the equipment be replaced if $i = 12\%$	10
	ii) At a central warehouse, vehicles arrive at the rate of 18 per hour and the arrival rate follows Poisson distribution. The uploading time of the vehicles follows exponential distribution and the unloading rate is 6 vehicles per hour. There are 4 unloading crews. Find the following: a) P_0 and P_3 b) L_q, L_s, W_q, W_s	10
	iii) A harbour has single dock to upload the container from the incoming ships. The arrival rate of ships at the harbour follows poisson distribution and the unloading time for the ships exponential (negative) distribution and hence, the service rate also follows Poisson distribution. The arrival rate and the service rate are 8 ships per week and 14 ships per week, respectively. Find the following a) Utilization of the dock b) Average number of waiting ships in the queue. c) Average number of waiting ships in the system. d) Average waiting time per ships in the queue. e) Average waiting time per ship in the system.	10